Getting urgent requests to the right people, fast

AI process redesign case study: resident service requests in a 450-bed residential building

Bushra Faruq Mujawar

Executive summary

1 Problem

Every request is read by a person before anyone knows it is urgent. Urgent emails wait a median 78 minutes; 18% of requests go to the wrong team.

2 Redesign

An AI agent reads each request on arrival and proposes category, priority and owner. A coordinator decides. Safety issues alert the duty manager at once.

3 Value

Coordinator effort falls from 156 to 64 hours a month. Net EUR 26.8k a year on an EUR 11k build; payback about 5 months.

Recommendation: approve a two-week shadow trial. The keyword prototype fails the safety gate (P1 recall 69% vs 95% needed), so go-live depends on the LLM version passing the same test.

Urgency is only discovered when someone reads the inbox

78 min

median wait before an urgent email request is triaged

64%

of urgent inbox requests wait over an hour

18%

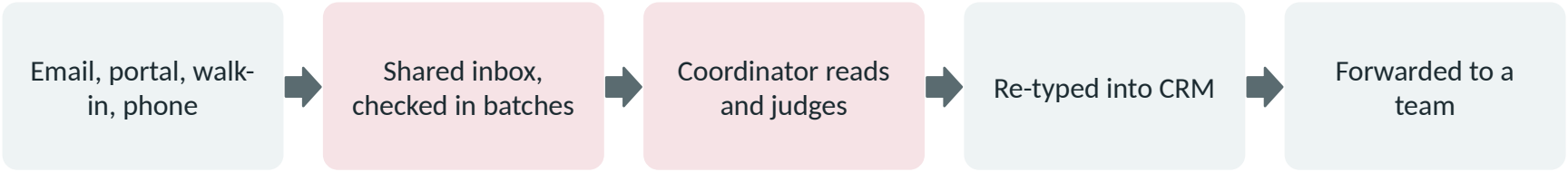
of requests go to the wrong team first time

156 h

coordinator time a month on triage, rework and chasers

Six months, 7,894 requests (synthetic). 70% arrive by email or portal; 30% arrive outside desk hours, when the inbox is not worked at all. Also: 25% of priorities are inconsistent and 29% of residents chase for an update.

Root cause: the first read of every request is manual



- **Batches and nights:** Inbox worked every ~2 hours, not at all 20:00-08:00
- **No shared rules:** Category and priority depend on who is on shift
- **Silent hand-offs:** No acknowledgement, so 29% of residents chase

5 whys, in one line

A ceiling leak waited 80 minutes because nobody knew it was urgent until a person read it, and nobody reads the inbox continuously. This first read is the step AI can take over.

Eight opportunities scored; three form Wave 1

Opportunity	Value	Feas.	Data	Risk*	Adopt.	Score	Decision
Instant acknowledgement + response time	3	5	5	5	5	**4.40**	Wave 1
AI triage & routing, human approval	5	4	4	4	4	**4.30**	Wave 1
Automated weekly KPI report	3	5	4	5	4	**4.10**	Wave 1
Duplicate / same-issue detection	3	3	3	4	4	**3.35**	Wave 2
Predictive maintenance	4	2	1	4	3	**3.00**	Wave 2
Sentiment escalation	2	4	3	3	3	**2.90**	Not now
Resident FAQ assistant	3	3	2	3	3	**2.85**	Not now
Autonomous resolution	4	2	2	1	2	**2.40**	Not now

Weights: value 30%, feasibility 20%, data readiness 15%, risk 20%, adoption 15%. *Risk scored 5 = low risk. Wave 1 is one product: triage creates the structured data the other two use.

Future state: the agent proposes, a person decides



Shares are the prototype's review-mode mix on the 80-request test set.

Controls built into the design

Human approval

No request is routed without a coordinator in release 1; overrides are logged and reviewed weekly.

Safety bypass + rule net

Safety and P1 skip the queue. Keyword safety signals can add an escalation on top of the model, never remove one.

Privacy by design

Emails, phones, IBANs, card numbers and Eircodes are removed before the model sees text or anything is logged.

Fail safe

Model output is validated against a schema; any failure falls back to rules, so no request is dropped.

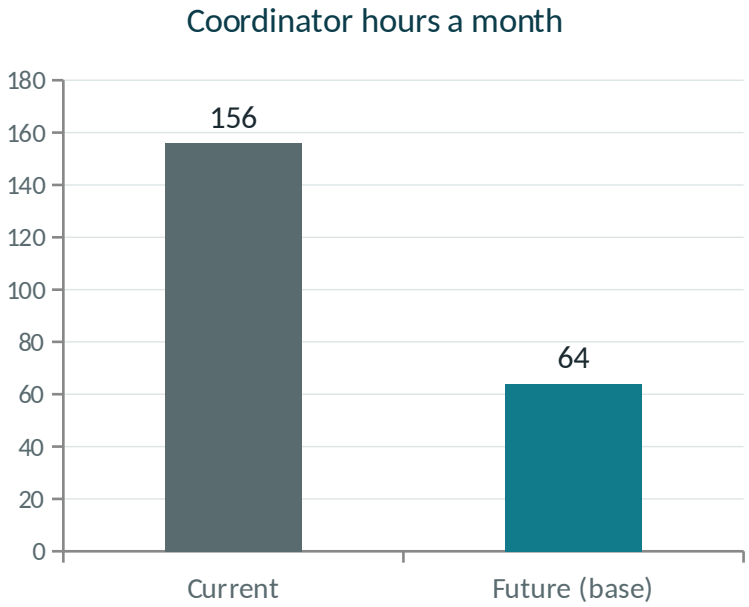
Audit trail

Every proposal, decision and override is written to a log for accountability and improvement.

Go-live gate

Pilot only if P1 recall and quick-approve precision are both at least 95% on the test set.

Value case: 0.6 of a coordinator back each month



Scenario	Net / year	Payback
Low (60%)	EUR 13,600	9.7 months
Base	**EUR 26,800**	**4.9 months**
High (125%)	EUR 35,000	3.8 months

- Build EUR 11,000 (20 days); run about EUR 6,200 a year.
- Saving is capacity, not headcount: time moves from admin to residents.
- The case fails only if coordinators don't trust proposals and re-read everything, hence the precision gate.

Prototype routes well but misses urgency

88.7%

Category accuracy

98.3%

Quick-approve precision

Gate 95%: pass

69.2%

P1 (urgent) recall

Gate 95%: fail

72.7%

Safety escalation recall

Gate 95%: fail

What the misses show

- Keywords find what a request is about, not how bad it is: "toilet overflowing onto the floor" was routed right but marked routine.
- The riskiest cases use ordinary words: "a man I don't know followed me in through the main door".
- Next: run the LLM classifier, with the rule safety net kept, through the same 80-request test.

Roadmap and decision needed

Weeks 1-2

Evaluate and shadow

LLM classifier through the test gate. Agent proposes in the background; coordinators work as today. Measure the real baseline.

Weeks 3-8

Pilot

Email and portal only, one building. Weekly override review; KPIs reported weekly.

Month 3+

Scale and Wave 2

All channels. Add duplicate detection; collect data for predictive maintenance.

Decision: approve two weeks of evaluation and shadow running (about 6 days of effort, no change for residents). Go or no-go on the pilot is set by the safety gate, not by opinion.